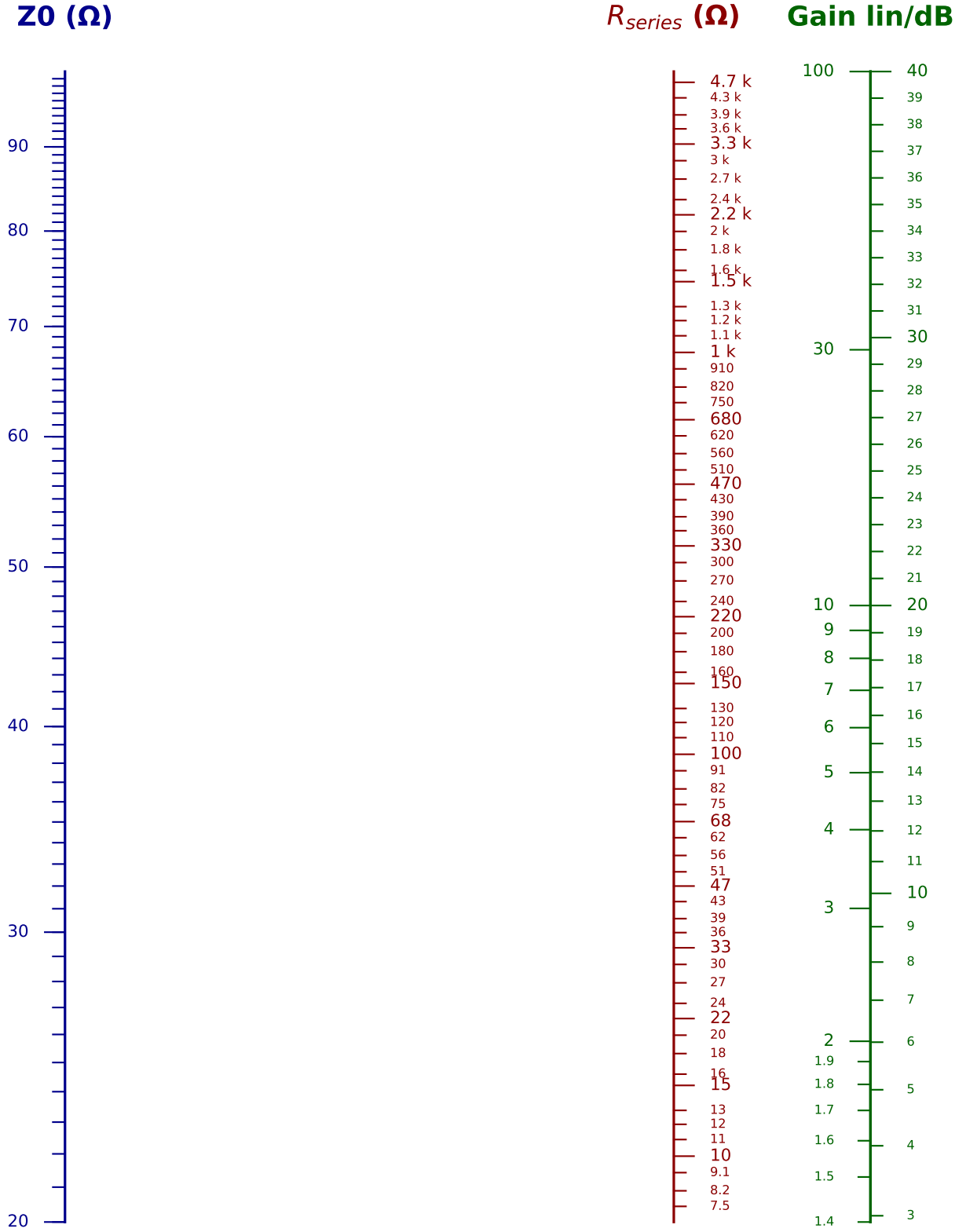
# Pi Attenuator Calculator $R_{series}$




$$R_{series} = \frac{Z_0}{2}\left(A - \frac{1}{A}\right)$$

$$A = 10^{\frac{dB}{20}}$$